

THANUJ KUMAR MALIGI

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Professional Summary

Data Science Specialist and Artificial Intelligence undergraduate with hands-on experience building machine learning pipelines, deep learning models, computer vision systems, and predictive data architectures. Skilled in Python, PyTorch, Scikit-Learn, OpenCV, SQL, and data processing tools (NumPy, Pandas, PySpark). Experienced in fine-tuning YOLOv11 object detection models, architecting UNet-based DDPM diffusion models on multi-spectral satellite data, and developing predictive data analytics. Published IEEE conference author and National Finalist at Code4Change Hackathon with a track record of delivering scalable AI/DS solutions.

Education

Amrita Vishwa Vidyapeetham, Bengaluru — B.Tech, Artificial Intelligence & Data Science 2023 – 2027
CGPA: **6.71 / 10** *Relevant Coursework: Machine Learning, Deep Learning, DBMS, Data Structures & Algorithms, Applied Statistics*

Narayana Junior College, Hyderabad — Intermediate (MPC) Aug 2021 – Mar 2023
Percentage: **81%**

Technical Skills

Data Science & ML: Machine Learning, Deep Learning, Neural Networks, Computer Vision, Predictive Modeling, Generative AI, Diffusion Models (DDPM), Object Detection, Model Fine-Tuning, Feature Engineering, Exploratory Data Analysis (EDA)

Libraries & Frameworks: PyTorch, Scikit-Learn, OpenCV, YOLOv8, YOLOv11, UNet, NLP, NumPy, Pandas, Matplotlib, Seaborn

Programming & Data Querying: Python, SQL (MySQL, PostgreSQL, Snowflake), Java, C, TypeScript, REST APIs

Big Data & Cloud Tools: PySpark, Apache Spark, Airflow, AWS (EC2, S3, Glue, Athena), Docker, Git, GitHub Actions, Linux

CS Fundamentals: Data Structures & Algorithms (DSA), DBMS, Object-Oriented Programming (OOP), Operating Systems, System Design

Data Science & AI Projects

Cloud Motion Prediction using Diffusion Models — Lead Machine Learning Engineer Apr 2026
PyTorch • Diffusion Models (DDPM) • UNet • Computer Vision • Predictive Modeling • Data Preprocessing

- Developed a Denoising Diffusion Probabilistic Model (DDPM) forecasting pipeline using INSAT-3DR multi-spectral satellite imagery for short-term meteorological prediction.
- Built automated data engineering and image preprocessing pipelines incorporating temporal stacking, patch extraction, and min-max normalization across 2,000+ sequential satellite image frames.
- Architected a UNet denoising neural network in PyTorch, optimizing spatial feature loss and improving forecasting accuracy.

Smart Traffic Operating System — Computer Vision & Data Analytics Lead Apr 2026
YOLOv11 • OpenCV • Python • Machine Learning • Multithreading • Automated Analytics • [GitHub](#)

- Fine-tuned a YOLOv11-S object detection model on the IISc UVH-26 Indian traffic dataset, achieving 28 FPS real-time vehicle classification while resolving detection gaps for auto-rickshaws and two-wheelers.
- Built a multithreaded simulation engine with occupancy-aware MLFQ signal scheduling algorithms, reducing average vehicle wait time by approximately 35%.
- Implemented real-time congestion estimation algorithms and automated CSV data logging pipelines to record traffic distribution analytics.

Full-Stack Placement Management System — Data Architect & SQL Engineer Dec 2025
SQL • MySQL • Data Modeling • REST APIs • Node.js • Express.js • [GitHub](#)

- Designed a 3NF normalized MySQL database schema with index-driven query optimization to support large-scale student placement analytics.
- Programmed automated candidate filtering and shortlisting algorithms based on academic CGPA and skill thresholds, reducing manual screening effort by 60%.
- Built secure RESTful APIs serving analytical data endpoints with JWT authentication and role-based access control.

Nexus: Network Resilience & Routing Simulator — Graph Data Algorithm Specialist Mar 2026
Graph Data Modeling • Algorithms (A, BFS, Tarjan's) • TypeScript • Next.js • [GitHub](#)*

- Engineered graph-based data models and path optimization algorithms (A*, BFS, Tarjan's) for dynamic network topology resilience analysis.
- Built an interactive telemetry dashboard for real-time failure injection analysis, achieving a 40% reduction in rerouting latency.

Achievements & Research

National Finalist, Code4Change Hackathon ISDM Goa — 2025–2026
Selected as 1 of 6 national finalists among 92 competing teams; designed fairness-aware data scheduling algorithms handling large-scale judicial datasets under a 24-hour build constraint.

Accepted Author, IEEE ICONAT Conference 2026
Authored *A Deadlock-Free Scheduling Framework for Adaptive Traffic Signal Control*; engineered concurrent real-time metadata streaming and analytical queue processing pipelines.

Certifications

AI Agent Hackathon — Certificate of Achievement Productspace (2025)
Deloitte Data Analytics Job Simulation — Certificate of Completion Forage (2024)